Before filming began, two impressive Caltech physics students filled notebooks with calculations about the Professor's equation. Elena Murchikova filled a clean, new notebook with calculations by grown-up Murph, calculations written with elegant calligraphy. Keith Matthews filled a beat up, old notebook with calculations by Professor Brand, in the more sloppy handwriting common for old guys like the Professor and me.

In the movie, grown-up Murph (played by Jessica Chastain) discusses the math in her notebook with the Professor (played by Michael Caine). Murchikova, an expert in quantum gravity and cosmology, was on set to advise Chastain about her dialog and notebook, and things she was to write on the blackboard. It was startling to see these two brilliant and beautiful women from very different worlds, both with bright red hair, huddled together.

As for me, I filled Professor Brand's blackboard with diagrams and mathematics (Figure 25.8, below), including the Professor's equation—THE equation—at Christopher Nolan's request, of course. And I took great pleasure in talking with Michael Caine (Figure 25.1), who seemed to view me as a sort of prototype for the Professor he was playing. And great pleasure in watching Chris, a master craftsman, mold the scenes he was filming into precisely the form he wanted.